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Stirring-based control strategy for microaerobic H_2S removal in sequencing batch anaerobic digesters

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ABSTRACT

Microaeration in fermenters is challenging due to the rapid increase of H_2S concentration upon the start of stirring, which makes biogas utilization less attractive. On these occasions, apparent H_2S production exceeds the ability of sulfur-oxidizing bacteria (SOB) to remove H_2S, which results in peak H_2S concentrations in biogas. This study shows that such peaks can be reduced by stirring optimization. For the first time, the effect of the stirring intensity on microaeration was evaluated in a lab-scale sequencing batch reactor and supported by a biokinetic mathematical model. Single stirring intensity (15, 25, 35, and 55 RPM) and gradually increasing stirring intensity (15 to 35 RPM) were compared. Despite achieving a stably low concentration of H_2S in the fermenter at low stirring speed, the accumulation of volatile fatty acids in liquid occurred. However, with gradually increasing stirring intensity, slower stirring at the beginning of the cycle led to the slower release of H_2S, thereby, winning more time for SOB to remove H_2S from biogas. H_2S peaks were reduced by this strategy and no accumulation of volatile fatty acids in the liquid phase was observed. Since microaeration is zero-waste chemical-free desulfurization method, its optimization and H_2S peaks reduction can help to expand its use.

1. Introduction

Hydrogen sulfide (H_2S) is an inevitable poisonous product of the anaerobic digestion of sulfur-rich substrates [12]. Burning biogas contaminated with H_2S shortens the lifetime of cogeneration units and increases operational costs due to the corrosion of concrete and steel structures. Also, when the biogas is upgraded to biomethane and injected into the natural gas distribution grid, H_2S concentrations must be negligible, typically below 5 ppm [5].

Complete desulfurization of biogas can be expensive, especially when treating biogas significantly contaminated with H_2S (thousands of ppm). Moreover, often used technologies such as alkaline wash or filtration on activated carbon require waste disposal or regeneration [15]. On the other hand, well-controlled microaeration can serve as a great zero-waste chemical-free method for biogas pre-treatment with subsequent post-treatment of residual H_2S concentrations on e.g., activated carbon filters.

Microaeration is a well-known biological method for the removal of H_2S from biogas directly in anaerobic digesters [12]. It is achieved by dosing small amount of oxygen or air into the anaerobic fermenter (typically into the headspace) to support the growth of sulfide oxidizing bacteria (SOB). SOB present in the biofilm colonizing the surfaces of reactor's headspace oxidize H_2S to harmless elemental sulfur [17]. Even though microaeration is commonly used in full-scale applications, there are still significant drawbacks associated with fermenters that are intermittently stirred, which may disqualify this biotechnology especially when biogas is upgraded to biomethane.

Intermittently mixed microaerated fermenters often experience peaks of H_2S concentrations in the headspace immediately after the start of stirring, resulting in insufficient biological oxidation. Similar peaks are observed when the substrate is dosed discontinuously [3]. This phenomenon is explained as a rapid release of H_2S from liquid after the start of the stirring. The typical response from operators to such peak production is the upregulation of dose air leading to biogas overloading

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with nitrogen and subsequent above-limit emissions of NO_x upon burning in a cogeneration unit [6]. Furthermore, H_2S peaks also increase the operational costs of biogas upgrading to biomethane. As a result, reducing the H_2S peaks and maintaining H_2S concentration on stably low levels without polluting or diluting the biogas by oxygen or air is a major challenge.

Intermittent stirring is often favorable choice for operators since it is robust, saves energy costs and can improve biogas production [20]. In reactors with intermittent stirring, biogas production as well as H_2S concentration in biogas rapidly increases after stirring starts. Stafford [21] tested different stirring intensities to prove, that biogas release after the start of mixing is higher than later during the mixing phase. However, the stirring regime can also significantly affect e.g. volatile fatty acid production, methane production, biogas composition, microbial population dynamics, or sheer stress [20]. Therefore, it is necessary to carefully adjust stirring intensity and monitor its effect on the entire anaerobic digestion process.

In our previous study, a new mathematical model of microaeration in SBR suggested that the intensity of stirring largely affected the H_2S release from liquid during operational cycle [3]. Therefore, we hypothesized that controlled release of H_2S from liquid due to proper stirring control can win more time for biological oxidation in the headspace and thereby significantly improve the efficiency of microaeration.

In this lab-scale study, we experimentally tested various stirring regimes of microaerated SBR, including different constant stirring intensities and gradually increased stirring intensity, to eliminate the H_2S concentration peaks in headspace. Consequently, we developed an optimal stirring-control strategy that assured a complete reduction of H_2S peaks.

2. Materials and methods

2.1. Set-up configuration

To study the effect of mixing intensity on microaeration efficiency, two laboratory acrylic glass reactors (experimental reactor R, and control reactor C) were build and operated for 360 days. Reactors were operated under mesophilic conditions (44 °C) and the volume of each reactor was 20 L including 13 L of sludge and 7 L of headspace. To maintain consistent influent composition, both reactors were fed by artificial substrate: cheese whey solution (40 g L^{-1}) with the addition of K_2SO_4 (1.8 g L^{-1}) to increase the amount of H_2S in the biogas. The homogenization of the sludge was ensured with a two-blade impeller. Pure oxygen was dosed continuously into the headspace of both reactors (4 mL h^{-1}) using a mass flow controller (Bronkhorst High-Tech B.V., Netherlands). Both reactors were operated as sequencing batch reactors (SBRs). The sedimentation phase was set to 50 min. The reaction phase involved stirring of the sludge and its length was adjusted from 3.5 up to 8 h based on experiments. After sedimentation, 100 mL of the substrate was dosed, and 100 mL of effluent was drained from the reactor simultaneously via a double-head peristaltic pump (Watson-Marlow Fluid Technology Group, UK).

The biogas production was measured by gas flow meter Milli-Gascounters (Dr.-Ing. RITTER Apparatebau GmbH & Co. KG, Germany) every 5 min. H_2S concentration in biogas was measured by an online H_2S sensor (H_2S/SP-10000, Membrapor, Switzerland) with automatic sampling equipment. The H_2S sensor was every 20 min washed with biogas for 5 min and then washed with nitrogen for 5 min. According to the sensor's specification datasheet, the resolution of the sensor is less than 3 ppm and the maximum zero shift is 10 ppm based on the environment temperature. This shift was recorded in the results, where the lowest measured value corresponds to exactly 10 ppm.

H_2S concentration, temperature, and biogas production measurement were controlled by an in-house program developed using LabVIEW 2012 software running on a CompactRIO system (National Instruments, US) (Fig. 1).

Sludge from a mesophilic (44 °C) anaerobic fermenter (Wastewater treatment plant Česká Lípa, Czech Republic) was used as inoculum. Total suspended solids and volatile suspended solids were 24.5 and 13.8 g L^{-1}, respectively.

2.2. Experimental design

After the inoculation, both reactors were operated without experiments for 9 months to reach steady state in both reactors and to allow sufficient time for biofilm growth on the headspace walls under microaerobic conditions. During this phase, stirring speed was 55RPM and the reaction phase of SBR took 3.5 h.

2.2.1. Stirring intensity experiments

To evaluate the effect of stirring intensity on H_2S release from liquid and, thus, microaeration efficiency, several experiments were designed. The effect of single-speed intensity on the one hand, and gradual increasing stirring intensity on the other hand were studied in R.

First, single-speed experiments were conducted to test four different stirring speeds: 15, 25, 35, and 55 RPM. During the experiment, the reaction phase was prolonged to from 3.5 to 8 h, to avoid acidification of the reactor, which may occur at lower stirring speeds (15 and 25 RPM) with reaction time 3.5 h due to the slower degradation of the substrate and subsequent accumulation of volatile fatty acids (VFA).

Secondly, gradual increasing of the stirring intensity within one feeding cycle was studied. At the beginning of the cycle, the stirrer speed was set to 15 RPM to reduce H_2S transfer rate from liquid to gas after stirring starts. After two hours, the speed increased to 35 RPM to increase H_2S transfer coefficient and to accelerate the release of the rest H_2S from liquid. The reaction time of these kinetic tests was 4.2 h (250 min).

For each single-speed experiment, the intensity was set, and the reactor was operated for two-weeks to see the impact of the intensity change in long-term. For gradual stirring speed, due to the manual switching of the speed requirement, 10 SBR cycles were tested with pause, where the speed was set to 35 RPM to see behavior of the system while changing stirring regimes.

2.2.2. Sulfide kinetics in the liquid

For each single-speed intensity test, kinetics of dissolved sulfide concentration in the liquid was measured. During the reaction phase, the sludge was sampled every hour and the concentration of sulfide was measured. In the 55-RPM test, the interval between the first two samples was shortened to 30 min to catch the peak of the sulfide concentration.

2.2.3. VFA monitoring

The overall conditions of the reactors were monitored by volatile fatty acids (VFA) concentration measurement since it is one of the most important reactor failure indicators [9]. VFA was measured 3 times in each intensity test to see potential accumulation.

2.3. Simulations

To support experimental results and to verify the functionality of the previously developed model [3], simulations in AQUASIM software [19] were done. Mathematical model includes 3 compartments: liquid phase, headspace, and biofilm. The equilibrium between adjacent compartments was described by Henrýs law. In the model, the faith of sulfur substrate was described by seven processes (the uptake of sulfate by SRB in the liquid, the chemical oxidation of H_2S in the headspace, the uptake of H_2S by SOB in biofilm, the uptake of S^0 by SOB, the decay of SOB, the decay of SRB, and washout of H_2S from the reactor with the produced biogas. For the simulations, the original mathematical model described in detail in Andreides et al. [3] was used and retained with the following minor modifications:

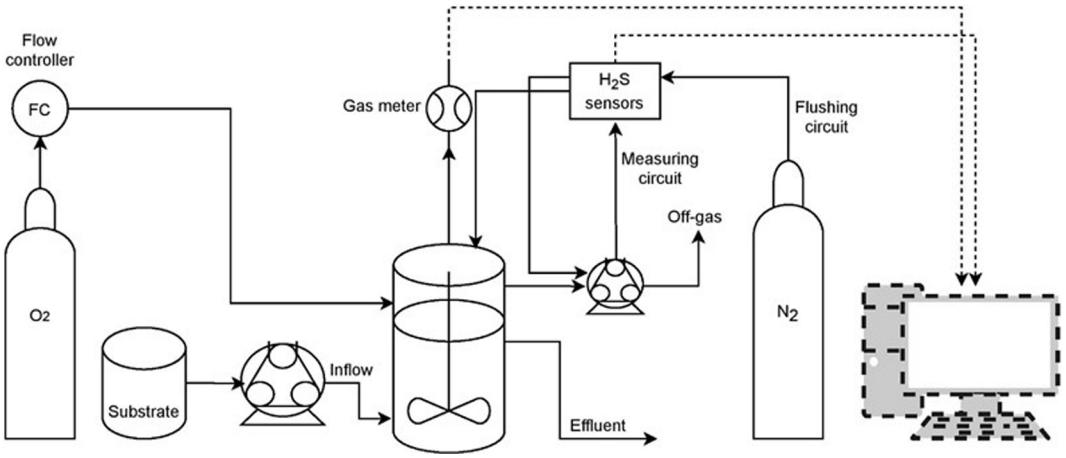


Fig. 1. The scheme of the reactor's set-up (solid line – mass flow; dashed line – information flow).

- The gas-to-biofilm mass transfer coefficient was increased (k_La_BF = 50 h^{-1}) since the biofilm area was bigger than in the previous study.
- The liquid-to-gas mass transfer coefficient k_La value for different stirring speed was adjusted to fit experimental data. As the stirring speed increased, the k_La value increased. The original k_La value for 55 RPM was kept at 20 h^{-1} as determined in the previous study [3].
- The concentration of sulfate was changed according to the real dose.

Previously, scenario analyses were done in AQUASIM (Fig. 2) by Andreides et al. [3]. It was shown that the change of the stirring speed expressed as a change of mass transfer coefficient k_La can significantly affect the H_2S concentration in the headspace, since the release of H_2S from liquid can be controlled and the peaks in H_2S concentration can be thus reduced. In this study, we verified this hypothesis and evaluate the real conditions of the system with various tested speed intensities in continuously operated lab-scale SBR.

2.4. Analytical methods

Every two weeks, the COD concentration of the sludge and effluent was measured according to the standard methods [4] and total suspended solids and volatile suspended solids were measured gravimetrically.

For the VFA concentration determination in the liquid phase, a gas chromatograph (GC 2010, Shimadzu, Japan) was used and the conditions were set according to Andreides et al. [2]: capillary column CP-Vax 58 (25 m length and 0.25 mm ID); flame ionization detector; the oven column temperature was set on 70 °C; the heating rate of 15 °C/min, until it reached 134 °C; the injector port and detector temperature

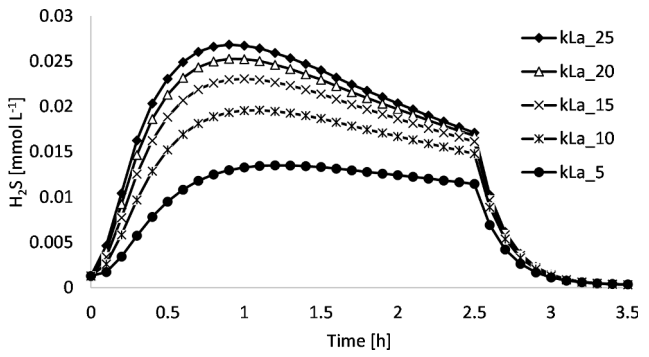


Fig. 2. H_2S concentration in the headspace; Scenario analyses of variable stirring intensities expressed as change of mass transfer coefficient kLa [h^{-1}]. This picture was taken from Andreides et al. (2021b).

270 °C and 300 °C, respectively. The time of the analysis was 5.27 min.

The analyses of biogas composition (methane, oxygen, nitrogen, and carbon dioxide) were performed three times a week at the beginning of the reactor's operation, and later every two weeks since no significant change was observed. The analyses were done as in Andreides et al. [3]: GC-TCD system (Shimadzu, Japan) with a stainless steel Hayesep D packed column (80/100) for biogas composition detection (carbon dioxide, methane, air) and a 5A molecular sieve (80/100) for air composition detection (oxygen, nitrogen): initial temperature of 150 °C, detector temperature of 185 °C, and isocratic analysis at 130 °C for both columns. Helium was used as the carrier gas, and the analysis time was 3 and 6 min for biogas and air detection, respectively.

2.5. Statistical analyses

For each tested stirring intensities, at least 4 operational cycles were performed and the average H_2S concentrations and its standard deviation were calculated. To assess the agreement between experimental and modeled data, root mean square error (RMSE) was calculated for each tested stirring intensity. The mathematical model was fitted to the experimental data to find the lowest RMSE.

3. Results and discussion

3.1. Overall performance of the reactors

pH and temperature were stable at 7.2 – 7.4 and 44 °C, respectively. During off experiment period, the concentration of total VFA in the reactors did not exceeded 75 mg L^{-1} (reaction cycle 3.5 h, 55 RPM). Thus, no accumulation of the acids was observed in this phase.

Biogas composition in reactors headspace was stable during entire reactors operation in both reactors (Table 1).

The concentration of H_2S in both reactors fluctuated between values close to 0 and approx. 200 ppm after the biofilm development and did not exceed 40 ppm, when the gradual stirring was applied. When the reactors were operated without microaeration, the concentration in R and C was 5800 ± 200 and 5600 ± 300 ppm, respectively. The H_2S removal efficiency of up to 97% was achieved, which is comparable to other studies of microaeration (e.g., [7,10,18].

The oxygen dosing points of R and C were clogged by elemental

Table 1
Average biogas composition.

Reactor	O_2 [%]	CH_4 [%]	CO_2 [%]
R	0.2 ± 0.2	59.9 ± 6.1	39.9 ± 6.1
C	0.2 ± 0.2	59.7 ± 6.4	40.0 ± 6.4

sulfur after 220 and 229 days, respectively, which was detected by increasing concentration of H_2S in the headspace. The clogging of the pipes is one of the most cited drawbacks of microaeration and regular control and cleaning should be scheduled for every microaerobic fermenter to avoid further problems [12]. The dosing points were cleaned up and no further clogging was observed.

3.2. Single stirring speed tests

3.2.1. Comparison of sulfide concentration in the liquid and headspace of the digester

Fig. 3 shows the comparison of sulfide kinetics in the liquid measured in prolonged cycles (8 h) with 4 different stirring speeds. Fig. 4 shows H_2S concentration in the headspace for different stirring speeds.

At **15 RPM**, sulfide concentration increased relatively slowly as H_2S transfer into biogas was slow. Therefore, H_2S oxidation rate in the headspace was sufficient and H_2S concentration was stable under 80 ppm. On the other hand, the two-blade impeller used for sludge homogenization did not ensure a proper homogenization at this low stirring speed, which was demonstrated in a 3D model of Abld et al. [1]. Due to the poor homogenization of the sludge, the degradation of organic substrate was significantly slower, and the reaction phase had to be prolonged to protect anaerobic culture against acidification. This is, indeed, not an attractive choice for operators of full-scale digesters. In addition, even after 8 h, sulfides were not fully depleted from the sludge and, thus, the concentration after the start of stirring was affected by residual sulfur compounds concentration from the previous cycle.

At **25 RPM**, the peak of H_2S concentration was around 100 ppm. When the duration of the reaction phase was kept at 8 h, the conditions of the anaerobic sludge were satisfactory, i.e., both VFA and sulfide was almost fully depleted from the liquid at the end of the cycle. H_2S concentration in the next cycle was influenced by minor residual sulfide concentration from the previous cycle.

At **35 RPM**, sulfide was depleted from the sludge after 4 h, but the H_2S concentration in the first hour of the reaction increased above 110 ppm. Even though the k_La was different with different stirring speeds, the highest H_2S concentrations during 25 and 35 RPM cycles were very similar. This was caused by the different initial sulfide concentration.

At **55 RPM**, sulfide was depleted within one hour. The highest concentrations were measured approx. 30 min after the start of mixing. As expected, H_2S concentration fluctuations were most significant with the highest H_2S concentrations, reaching almost 140 ppm.

When the reaction time decreased to 3.5 h, maximum H_2S concentration at both 35 and 55 RPM ranged around 200 ppm.

3.2.2. Monitoring VFA concentration

For every single-speed intensity (15, 25, 35, 55 RPM) VFA concentration in the reactor was monitored (Table 2) to overview conditions of anaerobic culture in the fermenter.

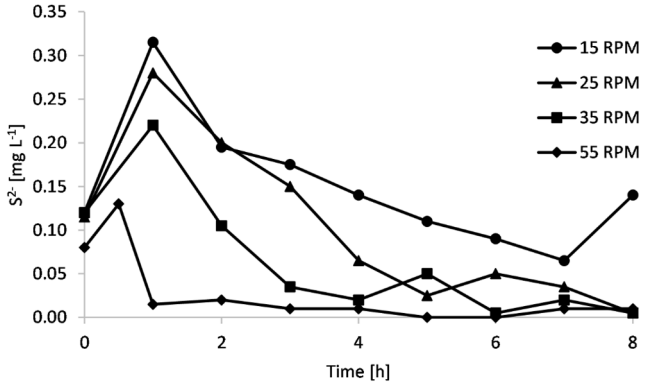


Fig. 3. Comparison of sulfide removal kinetics in the liquid phase of the microaerobic fermenter with different stirring speed.

During the 15-RPM experiment and reaction time 3.5 h, VFA concentration went up to 2 g L^{-1}. Even though the VFA inhibition limit depends on the buffering capacity of the sludge [16], anaerobic digestion in a full-scale application is considered to be stable when VFA concentration does not exceed 1 g L^{-1} [8]. So, the inhibitory concentration was exceeded with stirring speed 15 RPM, the pH of the reactor quickly dropped to 6.4 and as a result, feeding had to be stopped. The reaction phase was then prolonged to 8 h to avoid acidification. However, biogas production did not stop even after 8 h indicating that the substrate was not fully degraded yet. To maintain the complete degradation of the substrate, even longer time would be required for 15 and 25 RPM, which resulted from poor homogenization of the sludge. Consequently, organic loading rate significantly decreased, which would, in the long term, lead to biomass loss. Therefore, the reaction time was not extended beyond 8 h. In all cases mentioned in Table 2, acetic acid was the dominant VFA representing up to 80% of all VFA produced.

It is obvious from Table 2, that the higher stirring speed, the lower VFA concentration at the end of the cycle. With 55 RPM and reaction time 8 h, VFA were almost fully consumed. This phenomenon was also shown by Lindmark et al. [13]. During an experiment with 150 RPM, VFA consumption was higher compared to 25 RPM and the accumulation of VFA did not occur. On the other hand, too intensive stirring speed can lead to a chronic failure of the reactor due to the disruption of bacterial structure. Kim et al. [11] operated SBR with stirring speed 300 RPM which led to a reduced rate of hydrolysis and acidification of the reactor.

Mixing speed of 35 RPM seemed to be the optimal choice for the reactor under study, since stable anaerobic digestion could be achieved with shorter time than with 15 and 25 RPM and the H_2S peaks were lower than for 55 RPM.

To sum up, low stirring speed (15 RPM) was favorable for the efficiency of microaeration, because of the slower H_2S release from the liquid and, thus the lowest H_2S concentration in headspace. However, sludge homogenization was not sufficient, the accumulation of VFA occurred and pH in the liquid phase start to decrease. Conversely, at high stirring speed (55 RPM), anaerobic digestion was stable, but greater peaks in H_2S concentration in headspace were observed, because of the quick desorption of sulfide from the liquid phase. Therefore, to achieve both peak reduction and stable anaerobic fermentation, we designed an experiment with a gradually increasing stirring intensity.

3.2.3. The effect of mixing speed on biogas production

Biogas production was also affected by the intensity of mixing. In Fig. 5, three consecutive cycles for each tested stirring intensity are compared in terms of biogas production kinetics and total biogas production in the individual cycles. In the cycles with higher stirring intensity (35, 55, and gradually increasing stirring intensity), the production of biogas per cycle was 1297 ± 53 mL. Biogas productions in the cycles with 15 and 25 RPM was 1155 ± 54 mL, which coincides with increased VFA concentration (up to 2 mg HAc L^{-1}) remaining in the liquid at the end of each cycle. Also, the release of biogas from liquid was much slower at 15 and 25 RPM compared to the higher stirring intensities. Both in cycles 35 and 55 RPM, there was a long phase at the end of each sample where almost no biogas was produced. The composition of biogas was stable in every tested intensity (Tab. 1).

3.3. Gradually increasing stirring intensity tests

Fig. 6 shows the comparison between H_2S concentration in the headspace during SBR cycles with gradual stirring (five cycles, white blocks) and single stirring speed (two cycles, grey blocks in the middle). When the gradually increasing stirring intensity (15 to 35 RPM) was applied, H_2S concentration was stable under 40 ppm, peak H_2S concentrations were reduced, and no VFA accumulation was observed.

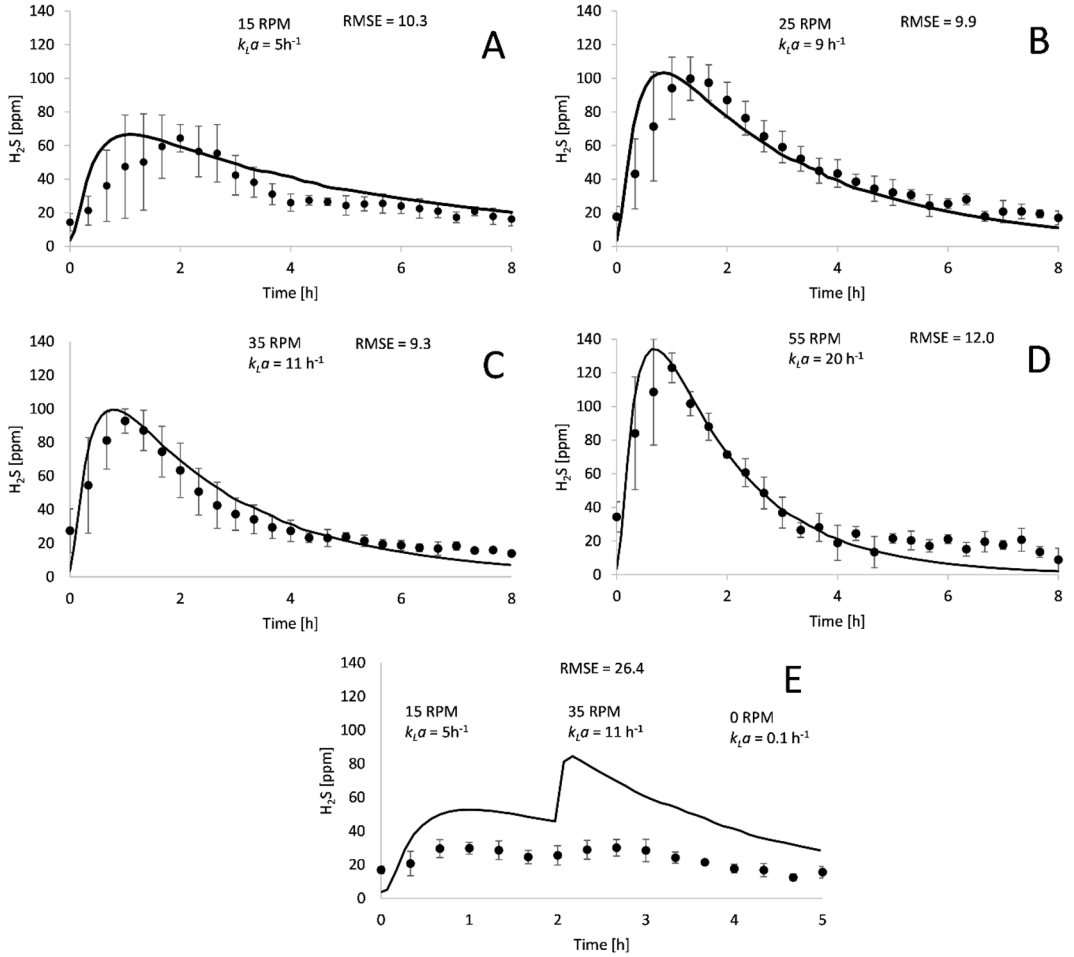


Fig. 4. Simulations of H_2S concentrations in the biogas produced in the microaerobic fermenter with different stirring regimes and speeds. Measured data – dots, mathematically modeled data – black lines. For every single-speed test, average values from 4 cycles were taken; standard deviations and RMSE were provided. Note, that graph E includes also the sedimentation phase from hour 4.17 to 5.

Table 2
Comparison of VFA concentration in the sludge at the end of the reaction cycles with different stirring speed (VFA concentration recalculated to concentration of acetic acid).

Stirrer speed	Reaction phase duration [h]	VFA concentration [mg HAc L^{-1}]
15 RPM	3.5	1924 ± 147
15 RPM	8.0	189 ± 48
25 RPM	8.0	89 ± 7
35 RPM	8.0	71 ± 26
55 RPM	8.0	10 ± 17
55 RPM	3.5	50 ± 26

When the stirring regime was switched again to single stirring (the 4th and 5th cycle in Fig. 6), peak H_2S concentrations were observed again.

Moreover, the reaction time could be decreased to 4 h and 10 min while assuring efficient VFA degradation. Here, the stirring intensity was changed manually and therefore only two speeds for the gradual increasing stirring intensity test were used. When stirring regimes switching is automatically controlled and a continuous increase of speed is used, the stirring control might be even more effective, and the reaction time could be even shorter.

We did not test gradual increase from 15 to 55 RPM due to the limitation of the equipment, but, based on the results from H_2S and sulfate concentration, there is an assumption, that the reaction time could be even shorter. In Fig. 6, it is visible that H_2S concentration did not drop to zero. This phenomenon is probably due to the limitation of the H_2S sensor mentioned in Chapter 2.1.

The peak concentrations of H_2S were lower with the gradual increase of stirring intensity compared to the single stirring speed test with 15 RPM. As was already mentioned, the residual concentration of sulfide in the liquid after the cycles with low stirring intensity can explain this phenomenon. On the other hand, when the sulfide depletion was reduced by increasing stirring intensity during the cycle, the peak concentrations at the beginning of the subsequent cycles (after the start of stirring) was reduced.

3.4. *Explaining the experimental results by mathematical model*

The simulation of microaeration with different stirring speeds using the mathematical model developed by Andreides et al. [3] is shown in Fig. 4A-D. The massive release of H_2S at the beginning of the reaction phase corresponds to gas release commonly observed in intermittently stirred fermenters [21]. Gas release from the liquid phase depends on the mass transfer coefficient, which is increased by increasing stirring intensity [14]. This assumption served as the basis for the mathematical simulation, which showed that H_2S release may decrease at lower stirring intensity (Fig. 2).

In the model developed by Andreides et al. [3] with changing biofilm area, only liquid-to-gas mass transfer coefficients k_La and substrate concentration was changed to fit the experimental data (Chapter 2.3.). The original k_La value derived for 55 RPM in the original mathematical model (20 h^{-1}) was changed for other stirring speeds (Fig. 4). Because of

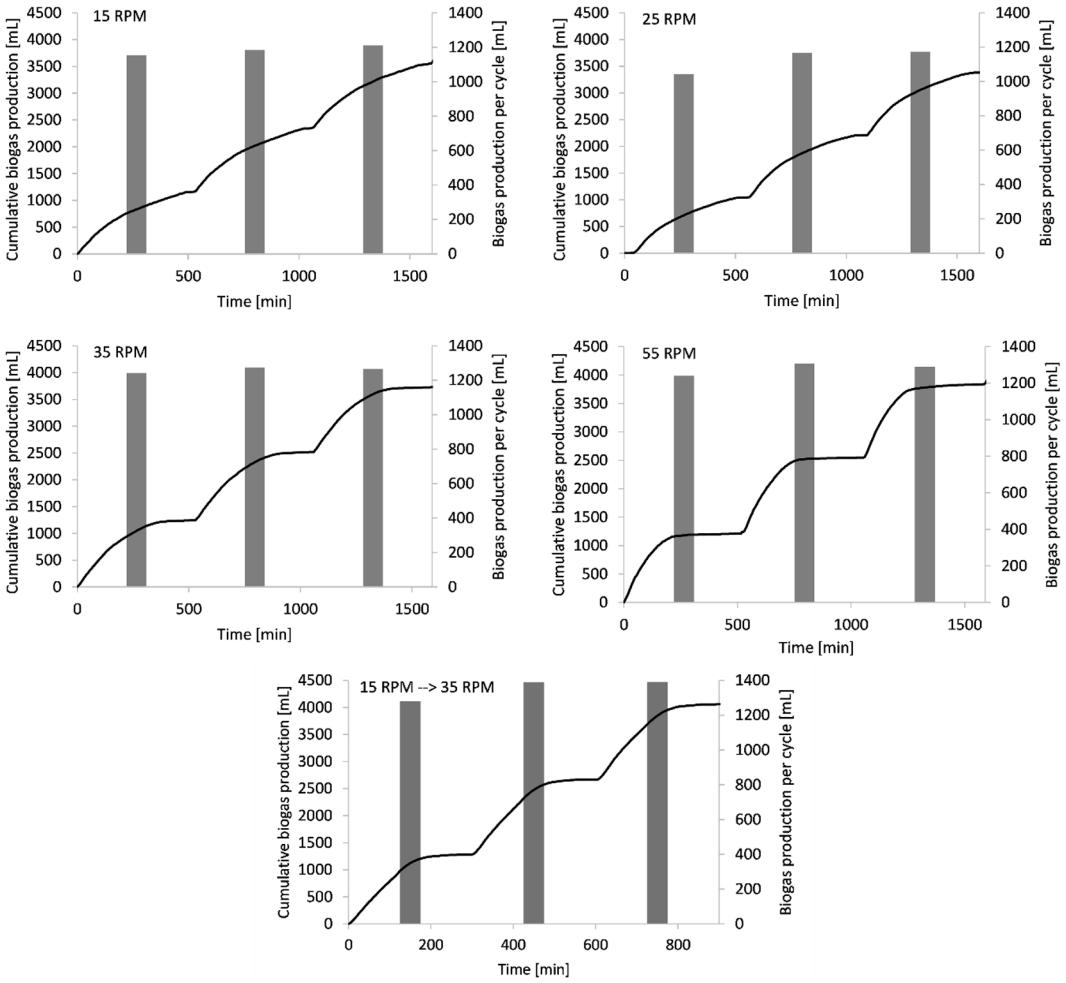


Fig. 5. Biogas production in microaerobic fermenter with different stirring regimes and speeds. Cumulative biogas production – black line, biogas production per cycle – gray columns. Note, that graph 15 → 35 has different range on the X-axis.

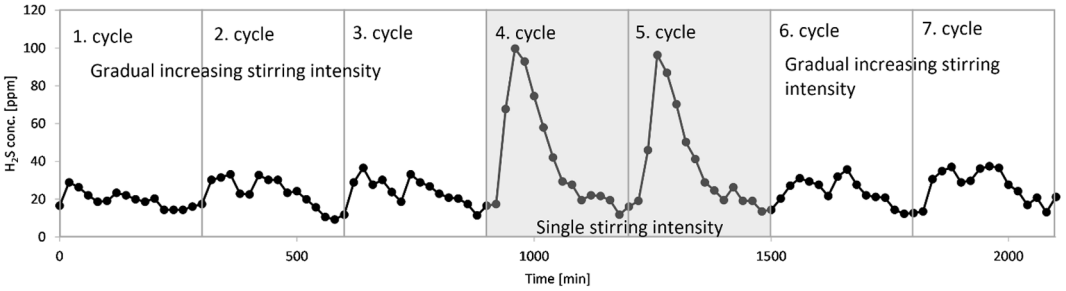


Fig. 6. Comparison between microaeration SBR cycles with gradually increasing stirring speed from 15 to 35 RPM (white rectangles) and single stirring speed 35 RPM (grey rectangles). The cycles were measured immediately after each other.

the residual sulfide concentration in the liquid at the end of cycles with 15 and 25 RPM (Fig. 3), modelled initial sulfate concentration for the experiments at 15 and 25 RPM was increased compared to 35 and 55 RPM.

For the simulation of the experiments with gradual stirring speed increase (4E), k_La values were used according to Fig. 4. In the first two hours, k_La was set to 7 h^{-1}, after two hours the value was increased to 12 h^{-1}. After 4 h and 10 min, the stirring stopped. In the mathematical model, the reactors are considered to be ideally stirred, so, the sedimentation was modeled as an instant drop of k_La to a minimum value. To avoid numerical problems in AQUASIM, k_La during sedimentation was set to 0.1 h^{-1}. The rapid drop in modeled data is not reflected in the measured data due to the limitations of the sensor. It is important to note, that AQUASIM assumes ideal homogenization in all modeled compartments (liquid phase, headspace, and biofilm), therefore, in such low concentrations, the results may differ.

Experimentally measured data followed the same trend as the data calculated from the model, and the differences were in the order of tens of ppm. Thus, the mathematical model satisfactorily described the fate of sulfur compounds in the microaerobic fermenter. Even though the experimental data showed a similar trend as was expected from the mathematical model, the real concentration of H_2S in biogas was even lower for the gradual increase of stirring intensity. Apparently, under these conditions, microaeration is affected by yet another factor (e.g.,

geometry of the system) that will require further investigation.

3.5. Applicability of the gradually increasing stirring intensity in full scale

Based on the results presented in this study, gradually increasing stirring intensity is an effective method for maintaining consistently low H_2S concentration in microaerobic SBRs and intermittently stirred and/or fed CSTRs. Reduced H_2S fluctuations in the biogas will lead to the extension of the applicability of microaeration in these reactors and to the mitigation of the side effects of H_2S presence in resulting biogas while burning it in cogeneration units.

Moreover, when pure oxygen is used, microaeration can be used for the pre-treatment when biomethanation is to be applied. In this study, thanks to microaeration, the concentration of H_2S was reduced from approx. 6000 ppm to approx. 40 ppm. Subsequent post-treatment on e.g., activated carbon or membranes would deal with only minor residual H_2S concentration, leading to a longer lifetime of chosen technology, and will provide important cost savings.

4. Conclusion

We experimentally verified a new operational strategy for efficient microaeration in anaerobic fermenters with intermittent mixing which was previously suggested based on mathematical modelling. The strategy was based on gradual increase of stirring speed during each feeding cycle of the fermenter. In this study, the optimal conditions were the slow start of the stirring (15 RPM) for two hours, and subsequent increase of stirring intensity to 35 RPM for another 2 h and 10 min. It was shown that such strategy reduced the maximum H_2S concentration in biogas from 6000 ppm (without microaeration) to less than 40 ppm compared to the peak concentrations of up to 120 ppm for RPM between 15 and 55. As explained by the mathematical model, the reduction of H_2S peaks resulted from the slowdown of H_2S release from liquid, which provided more time for biological oxidation in the headspace at the beginning of each SBR cycle.

Importantly, the strategy functioned without having any negative effect on anaerobic fermentation and did not require any extension of reaction time. Because the operational strategy proposed does not require additional equipment, it can be immediately implemented in industrial practice.

CRediT authorship contribution statement

M. Andreides: Writing – original draft, Conceptualization, Investigation, Methodology, Data curation, Formal analysis. **L. Pokorná-Krayzelová:** Supervision, Methodology. **J. Bartáčková:** Resources. **J. Bartáček:** Supervision, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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